

Steam methane reforming over a catalyst-coated wire with Joule heating: Heat and mass transfer study in catalyst-wire system

Dmitry Pashchenko^{ID}*, Viacheslav Papkov^{ID}

Department of Mechanical Engineering (Robotics) and MATEC, Guangdong Technion-Israel Institute of Technology, 241 Daxue Road, Guangdong, Shantou, 515063, China
Faculty of Mechanical Engineering, Technion - Israel Institute of Technology, Technion City, Haifa 3200003, Israel

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ABSTRACT

Steam methane reforming (SMR) driven by renewable electricity has emerged as a promising technology for low-carbon hydrogen production. This study focuses on SMR over a catalyst-coated wire heated by Joule heating, aiming to explore the complexities of heat and mass transfer within the catalyst-wire system. Numerical modeling utilizes to evaluate the effect of various parameters such as temperature (up to 1400 K), Reynolds number (up to 2000), and catalyst layer thickness (1 mm, 2 mm, and 5 mm), on energy and material balance. The maximal temperature wire was chosen as control parameters and volume heat flow from hot wire to catalyst layer was changed to change the maximal wire temperature. The numerical model was validated against experimental data. The results reveal that a 1-mm-thick catalyst layer behaves as thermally thin ($Bi < 0.1$), with catalyst temperature closely resembling wire temperature. However, increasing the catalyst layer thickness to 5 mm leads to a notable rise in the temperature difference between the catalyst layer and the wire. Moreover, thicker catalyst layers of 5 mm suffer from intra-layer diffusion limitations, which can reduce overall process efficiency. It was established that inside catalyst layer for a temperature range above 1200 K the complete methane conversion close to equilibrium value can be achieved.

1. Introduction

The utilization of renewable energy sources has become a crucial aspect in addressing global climate change and reducing greenhouse gas emissions. One promising application of renewable electricity is in the field of chemical processes, particularly in steam methane reforming (SMR) [1,2]. SMR is a well-established industrial process for producing hydrogen and synthesis gas from methane, which involves the reaction between methane and steam over a catalyst at elevated temperatures [3,4]. Traditionally, this process relies heavily on fossil fuels as an energy source, but with the constant improvement of renewable electricity generation technologies, there is potential to significantly reduce the carbon footprint associated with SMR [5–8].

Traditional SMR approaches typically rely on external heating sources [9–11]. Wismann et al. proposed to use renewable electricity as a heat source for steam methane reforming [12]. This paper introduces a groundbreaking method for streamlining a foundational process-integrating an electrically heated catalytic structure directly into a steam methane reforming (SMR) reactor for hydrogen production. This close coupling of the electric heat source and the reaction site enhances the reaction's proximity to thermal equilibrium, boosts

catalyst usage, and curbs undesirable byproducts. The compact design, featuring minimal characteristic length scales, facilitates significantly smaller reactor sizes—potentially up to 100 times smaller than existing reformer platforms. The authors concluded that electrification of SMR opens doors for innovative reactor designs potentially resulting in a substantial reduction of nearly 1% of global CO₂ emissions if implemented globally.

Lu and Nikrityuk studied the steam methane reforming process in a packed bed driven by the Joule heating [13]. This paper introduces a novel particle unresolved 3D-1D DEM-CFD-based model for predicting steam methane reforming (SMR) driven by direct electrical current. The model utilizes metal particles heated by Joule effect to sustain endothermic reactions, incorporating LHHW kinetics and six gaseous species. Key features include calculating 3D electric field distributions and comparing model predictions with chemical equilibrium-based alternatives.

The traditional approach of the use of Joule heating in the packed beds suffers from low heat transfer efficiency from hot heat source to catalyst particles [12,14,15]. There are many attempts to increase the efficiency of such packed beds. George et al. demonstrated that using

* Corresponding author at: Faculty of Mechanical Engineering, Technion - Israel Institute of Technology, Technion City, Haifa 3200003, Israel.
E-mail address: dmitry.pa@technion.ac.il (D. Pashchenko).

metallic foam substrates can enhance thermal conductivity significantly compared to traditional packed beds [16]. Zheng et al. overviewed the recent papers on Joule heating including a hybrid heating approach combining Joule heating with microwave assistance, achieving more uniform temperature distribution [17]. Recent work by Para-Cabrera's group showed that structured catalyst architectures with engineered thermal pathways can significantly reduce thermal gradients [18].

However, recent advancements have explored alternative heating mechanisms, including Joule heating of catalyst-coated surfaces (e.g. plate micro-reformers, catalyst coated wires, etc.), aiming to enhance efficiency and sustainability [19–21].

The use of electricity to heat catalyst-coated microreformers for steam methane reforming (SMR) has garnered significant attention due to its potential to enhance heat transfer efficiency. Several studies have explored this approach, highlighting its advantages over traditional heating methods [22,23].

The study by Burra et al. [24] examined the performance of a microreformer with a catalyst-coated nickel foam substrate heated by resistive heating. They reported that the direct application of electrical current resulted in rapid and uniform heating of the catalyst layer, leading to improved heat transfer efficiency compared to conventional conduction or convection-based heating methods. This enhancement was attributed to the minimization of thermal resistance between the heat source and the catalyst, allowing for better control over reaction conditions.

Another study provided by Pourali et al. [25], who developed a numerical model to simulate the heat and mass transfer processes in an electrically heated microreformer. Their findings suggest that the high surface-to-volume ratio inherent in microreformers, combined with the direct electrical heating mechanism, promotes efficient heat transfer from the heat source to the catalyst layer. This results in lower energy losses and higher overall process efficiency.

Finally, Yadav and colleagues [26] published a review to compare the performance of electrically heated microreformers with those heated by conventional methods. Their results confirmed that the former exhibited significantly higher heat transfer efficiencies, leading to faster start-up times, improved reaction rates, and reduced energy consumption.

These studies collectively demonstrate that the use of electricity to heat catalyst-coated surfaces offers distinct advantages in terms of heat transfer efficiency, reaction control, and overall process performance.

Recently, the catalyst-coated wires driven by Joule heating have found a great interest among researchers and engineers due to the potential efficiency. Ledesma et al. [27] investigated the feasibility of using renewable electricity to drive SMR reactions over catalyst-coated coil. They demonstrated that direct electrical heating could efficiently initiate and sustain the endothermic SMR reaction, providing precise control over temperature distribution within the reactor. Their findings highlighted the potential for reduced energy consumption and enhanced process efficiency compared to traditional methods.

Zheng et al. [17] reported a comprehensive study focusing on the heat and mass transfer characteristics within the catalyst-wire system. Their review revealed intricate interactions between electrical input, thermal gradients, and catalytic activity, emphasizing the importance of understanding these dynamics for optimal system performance. They proposed a novel model integrating both theoretical and experimental data, offering valuable insights into the optimization of such systems.

Furthermore, Kim et al. [28] extended the scope of investigation by exploring the effects of different catalyst and wire materials and coating thicknesses on the overall efficiency of SMR over Joule-heated wires. Their results indicated that certain combinations of catalyst and wire material and thickness could significantly improve hydrogen yield and minimize energy losses. This work underscores the need for careful selection and engineering of catalyst properties to maximize the benefits of Joule heating technology.

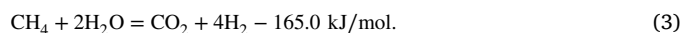
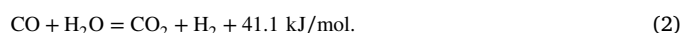
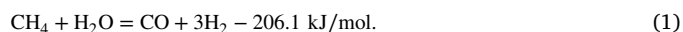
Overall, the literature reviewed here highlights the growing interest and progress in utilizing Joule heating of catalyst-coated wires for SMR. These studies collectively demonstrate the potential for enhancing process efficiency, reducing environmental impact, and improving control over reaction conditions. Future research should continue to explore optimized designs and integration strategies to further advance this technology.

The present paper focuses on the use of renewable electricity to power SMR via Joule heating of catalyst-coated wires. By directly applying electric current through these wires, it is possible to achieve rapid and efficient heating of the catalyst surface, thereby initiating and sustaining the endothermic SMR reaction. This approach not only reduces reliance on conventional fossil fuel-based heating methods but also offers precise control over temperature distribution within the reactor, potentially leading to improved process efficiency and product selectivity.

The primary objective of this work is to investigate the heat and mass transfer within the catalyst-wire system during SMR. Understanding these phenomena is essential for optimizing the design and operation of such systems, ensuring maximum conversion rates while minimizing energy consumption. Through detailed modeling and experimental validation, we aim to elucidate the complex interplay between electrical input, thermal gradients, and catalytic activity, ultimately contributing to the development of more sustainable and efficient hydrogen production technologies.

2. Numerical simulation

The steam methane reforming process involves various chemical reactions to form CO, CO₂, H₂, and C. However, there is sufficient information in the literature [29–31], and according to thermodynamic analysis at moderately high temperatures and a steam-to-methane ratio above 1.0 [32], only three chemical reactions have a notable effect on material and energy balances. Therefore, in this study, the following chemical reactions were considered:



2.1. Computational domain and mesh

The numerical simulation was performed for 2D computational domain as shown in Fig. 1a. The stream of methane, steam and nitrogen passes around a cylindrical hot wire, on the surface of which a layer of catalyst is coated. The diameter of the cylindrical hot wire is constant for all the simulated cases. The thickness of catalyst layer was varied for the different cases. In this paper, we considered three catalyst layer thicknesses: 1 mm, 2 mm and 5 mm. The catalyst layer is industrial Ni-based catalyst produced by ICI Katalco. The diameter of a wire is 10 mm.

The computational mesh for this 2D computational domain is presented in Fig. 1b. The mesh is structured. In the flow region, the inflation is added to obtain stable solution near wall region. The catalyst layer and wire contains 10 rows of cells per 1 mm in radial direction. The total number of mesh elements was determined based mesh independence study. Temperature of reaction mixture after cylinder was chosen as a control parameter.

2.2. Governing equations

The equations of mass, momentum, and energy conservation, coupled with chemical kinetics, diffusion, and porous medium models, were solved via ANSYS Fluent for steady-state conditions [33–35].

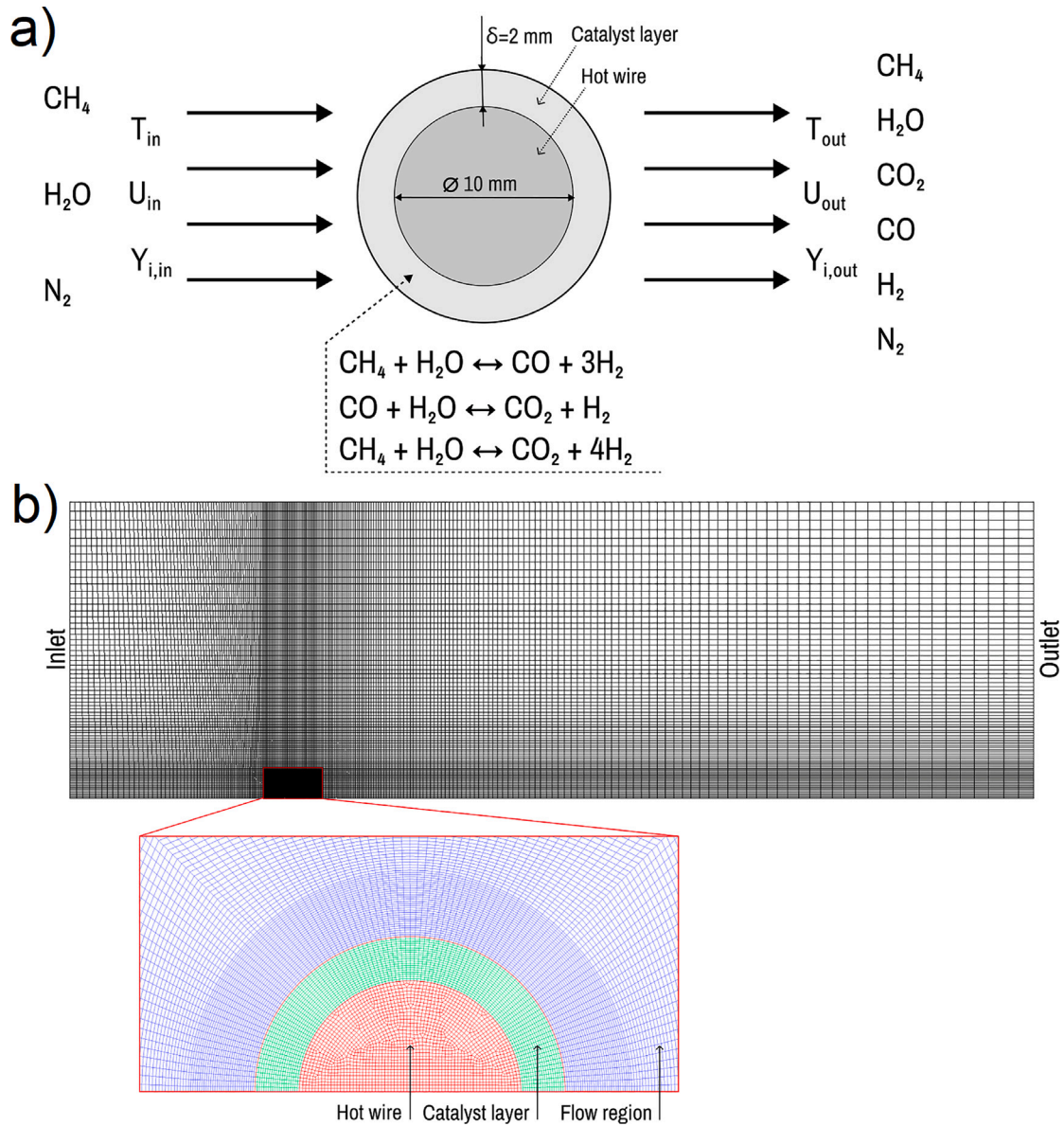


Fig. 1. The computational domain (a) and mesh structure for the analyzed problem (b).

Mass, momentum, energy conservation

The mass and momentum equations have the following form [36]:

$$\nabla \cdot (\rho \vec{v}) = 0; \quad \nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla \cdot p + \nabla \cdot (\bar{\bar{\tau}}) + \rho \vec{g} + \vec{F} + S_i \quad (4)$$

where p – pressure; $\rho \vec{g}$ – gravitational body force; $\vec{F}$ – external body forces; S_i – momentum source term for porous medium.

Stress tensor $\bar{\bar{\tau}}$ in Eq. (4) can be expressed as:

$$\bar{\bar{\tau}} = \mu \left[(\nabla \vec{v} + \vec{v}^T) - \frac{2}{3} \nabla \cdot \vec{v} I \right] \quad (5)$$

where μ – molecular viscosity; I – unit tensor, and the second term on the right hand side is the effect of volume dilation.

The energy equation is based on the law of energy conservation and has the following form:

$$\nabla \cdot \vec{v} (p + \rho \cdot h_f) = \nabla \cdot \left[k_{eff} \cdot \nabla T - \left(\sum h_j \cdot \vec{J}_j \right) + \bar{\bar{\tau}} \cdot \vec{v} \right] + S_f^h \quad (6)$$

where T – reaction mixture temperature; h_f – reaction mixture enthalpy; $\vec{J}_j$ – diffusion flux of j element; k_{eff} – thermal conductivity; h_j – enthalpy of j component; S_f^h – the source of energy due to chemical reactions.

The effective thermal conductivity of the porous medium, k_{eff} , is defined as follows:

$$k_{eff} = \epsilon_p k_f + (1 - \epsilon_p) k_s \quad (7)$$

where k_f , k_s – fluid phase and solid medium thermal conductivity, respectively; ϵ_p – porosity of catalytic particle.

The P1 radiation model is used to account for the radiative heat flux [37].

The transition SST turbulence model is employed to solve the RANS equations [38].

Porous medium

The porous medium of the catalyst layer is defined by the addition of a momentum source term (S_i) to Eq. (4). The actual pore size of the catalyst particles ranges from 10 to 100 nm. For such small pore sizes, the momentum transported into the particle is negligible. Therefore, the transport phenomena are defined by a permeability (α):

$$S_i = -\frac{\mu}{\alpha} v_i \quad (8)$$

The permeability (α) depends on pore size, tortuosity, etc. The permeability of the Ni-Al₂O₃ catalyst is set at $2.8 \cdot 10^{-14} \text{ m}^2$ [39].

Diffusion

The effective diffusion of i th species based on Bosanquet approximation [40]:

$$D_i^{eff} = \left(\frac{1}{D_{m,i}^{eff}} + \frac{1}{D_{i,k}^{eff}} \right)^{-1} \quad (9)$$

where $D_{m,i}^{eff}$ – effective ordinary diffusivity; $D_{i,k}^{eff}$ – effective Knudsen diffusivity.

Effective ordinary diffusivity ($D_{m,i}^{eff}$) is defined from:

$$D_{m,i}^{eff} = \frac{\epsilon}{\tau} \frac{1 - x_i}{\sum \frac{x_i}{D_{ij}}} \quad (10)$$

where ϵ – porosity; τ – tortuosity; x_i – molar fraction of i th species; D_{ij} – binary diffusivity for i - j th pair.

The binary diffusivity (D_{ij}) is calculated from:

$$D_{ij} = \frac{0.0143T^{1.75}}{p_i M^{0.5} \left(V_i^{1/3} + V_j^{1/3} \right)^2} \quad (11)$$

where $V_{i,j}$ – diffusion volume for i, j th species (from [41]: $H_2=7.07$; $H_2O=12.7$; $CO_2=26$; $CO=18.9$; $CH_4=16.8$)

The effective Knudsen diffusivity is determined from:

$$D_{i,k}^{eff} = \frac{\epsilon}{\tau} 48.5 d_p \left(\frac{T}{M_i} \right)^{0.5} \quad (12)$$

where d_p – mean pore diameter.

According to micropore-macropore model from Wakao and Smith's paper [42], a correlation between porosity and tortuosity for Al_2O_3 catalyst support can be expressed as:

$$\epsilon \approx \frac{1}{\tau} \quad (13)$$

Kinetic model

The reactions (1)–(3) occur in porous catalysts, and their reaction rates are determined based on Hou's kinetic model, which is widely used for numerical simulations of steam methane reforming [30]:

$$R_1 = \frac{k_1(p_{CH_4}p_{H_2O}^{0.5} - p_{H_2}^3p_{CO}/K_1p_{H_2O}^{0.5})}{p_{H_2}^{1.25}} \times \frac{1}{DEN^2} \quad (14)$$

$$R_2 = \frac{k_2(p_{CO}p_{H_2O}^{0.5} - p_{H_2}p_{CO_2}/K_2p_{H_2O}^{0.5})}{p_{H_2}^{0.5}} \times \frac{1}{DEN^2} \quad (15)$$

$$R_3 = \frac{k_3(p_{CH_4}p_{H_2O} - p_{H_2}^4p_{CO_2}/K_3p_{H_2O})}{p_{H_2}^{3.5}} \times \frac{1}{DEN^2} \quad (16)$$

$$DEN = 1 + K_{CO} \cdot p_{CO} + K_H \cdot p_{H_2}^{0.5} + K_{H_2O} \cdot p_{H_2O}/p_{H_2} \quad (17)$$

The expressions for coefficients in (14)–(16) are presented in Table 1::

To implement both chemical kinetics model and diffusion model in the Fluent solver, each model was converted into a User-Defined Function (UDF) written in C++ and uploaded to the Fluent solver.

Biot number

To assess whether the catalyst layer can be approximated as thermally thin (i.e., where temperature gradients are negligible), the Biot number (Bi) was evaluated. The Bi compares the internal thermal resistance of the catalyst layer to the external convective heat transfer resistance at its surface:

$$Bi = \frac{h \cdot L}{\lambda} \quad (18)$$

where h – convective heat transfer coefficient at the catalyst surface, L – characteristic length (thickness of the catalyst layer), λ – thermal conductivity of the catalyst material.

Table 1

The coefficients for the equations of kinetic model.

Kinetics		
k_1	=	$5.922 \cdot 10^8 \exp\left(\frac{-209200}{R_g T}\right)$
k_2	=	$6.028 \cdot 10^{-4} \exp\left(\frac{-15400}{R_g T}\right)$
k_3	=	$1.093 \cdot 10^3 \exp\left(\frac{-109400}{R_g T}\right)$
Adsorption		
K_{H_2O}	=	$9.251 \cdot \exp\left(\frac{-15900}{R_g T}\right)$
K_{H_2}	=	$5.68 \cdot 10^{-10} \exp\left(\frac{93400}{R_g T}\right)$
K_{CO}	=	$5.127 \times 10^{-13} \exp\left(\frac{140000}{R_g T}\right)$
Equilibrium		
K_1	=	$1.198 \cdot 10^{17} \exp\left(\frac{-26830}{T}\right)$
K_2	=	$1.767 \cdot 10^{-2} \exp\left(\frac{4400}{T}\right)$
K_3	=	$2.117 \cdot 10^{15} \exp\left(\frac{-22430}{T}\right)$

The heat transfer coefficient was determined based on the authors' previously published works [34,35,43]. The coefficient depends on both design and operational parameters of the system. The catalyst layer thickness varies in the range of 0.001 to 0.005 m. The effective thermal conductivity of the $Ni-\alpha Al_2O_3$ catalyst was taken as 18.9 W/(m·K) based on our experimental tests [44].

2.3. Boundary conditions and solution procedure

The velocity inlet boundary conditions are set as follows: $u_x = u$, $v = 0$, $p_g = 0$. At the outlet, pressure outlet boundary conditions are applied. The inlet reaction mixture temperature was 500 K. The temperature of hot wire was set as $T = 800, 1000, 1200$ and 1400 K. The operating pressure is 120 kPa. Reynolds number was varied as 50, 100, 500, 1000, and 2000. The inlet mass fraction is 0.3/0.6/0.1 of $CH_4/H_2O/N_2$.

The parameters of the Fluent solvers are as follows:

- Solver Type: Pressure-Based;
- Time: Steady State;
- Solution Scheme: SIMPLE;
- Spatial Discretization:
 - Gradient-Gauss Node Based;
 - Pressure: standard;
 - Momentum, Energy, Turbulence, Species: Third-Order MUSCL;
- Controls for Species and Energy: 0.95;
- Residual:
 - Energy: 1e–06;
 - Others: 1e–05;
- Initialization: Standard;

An industrial Ni-based catalyst is used for catalyst layer. This catalyst consists of nickel oxide on a calcium aluminate support, typically containing 16% NiO, 1.5% SiO_3 and 0.05% SO_3 . The catalyst properties are as follows: surface area of 14.3 m²/g; skeletal density of 3190 kg/m³; porosity of 0.44. The catalysts are the same as those used in the experimental paper by Hou and Hughes [30].

The maximal wire temperature was chosen as a control parameter for a volume heat flux. We defined the enthalpy of chemical steam methane reforming process and change of enthalpy of reacting mixture

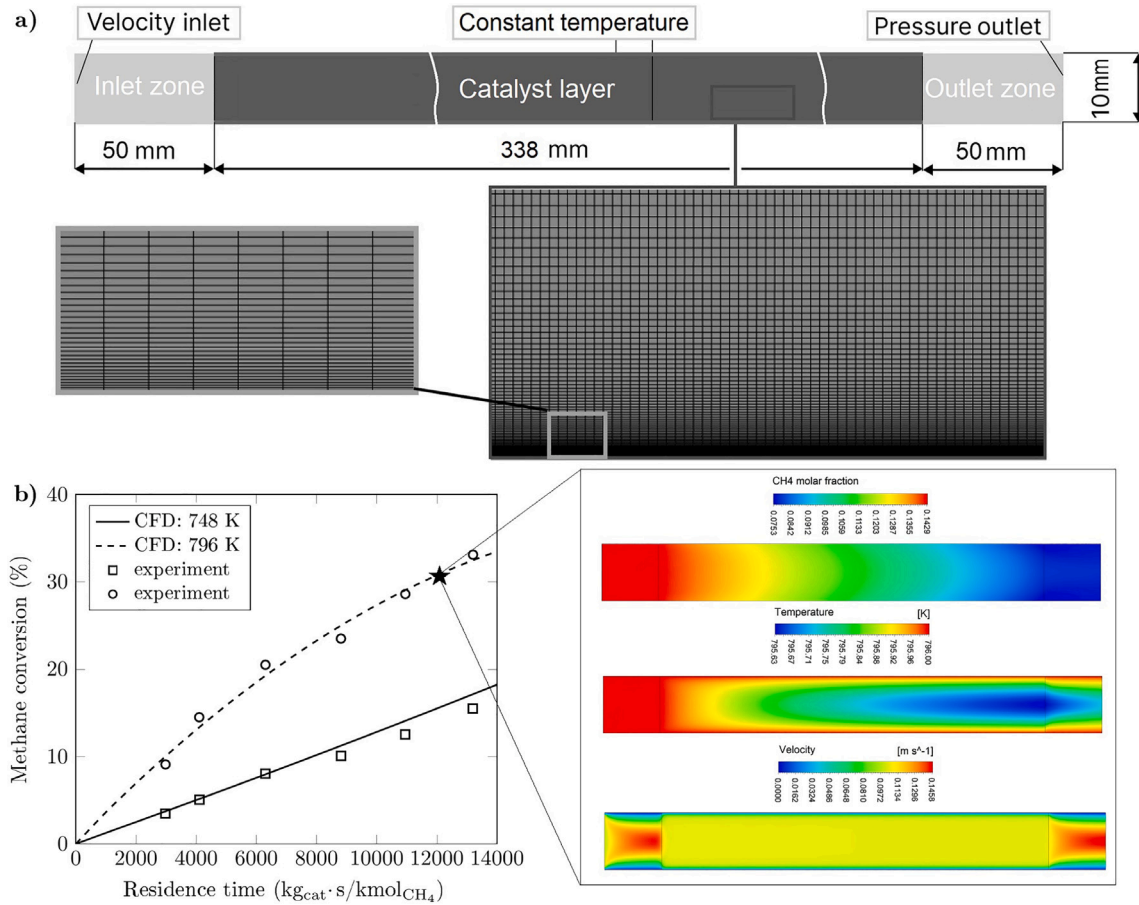


Fig. 2. Verification for the numerical model against experimental data: (a) computational domain and mesh; (b) Comparison of numerical results (dashed and solid lines) and experimental data (hollow points); contours of CH₄ molar fraction, temperature, and velocity for T=796 K and residence time of 12000 kg_{cat}·s/mol_{CH₄}.

per 1 m³ of wire. Therefore, the volume heat flux (Q''') was calculated as:

$$Q''' = (H_{SMR} + \Delta H_{mix}) / (\pi r^2), \quad (19)$$

where H_{SMR} – enthalpy of steam methane reforming process; ΔH_{mix} – change of reacting mixture enthalpy; r – wire radius.

The sum of the enthalpy of chemical steam methane reforming process and change of enthalpy of reacting mixture for 2D computational domain defines the heat flow per running meter of wire (Q'):

$$Q' = H_{SMR} + \Delta H_{mix}. \quad (20)$$

3. Results and discussion

3.1. Validation against experimental data

In this paper, the mathematical equations of model were numerically solved using ANSYS Fluent. To validate the model, the numerical results were compared with experimental data obtained by Hou and Hughes [30]. In our model, we used the kinetic model for steam methane reforming developed by Hou and Hughes [30]. Therefore, we performed a series of numerical calculations using the equations from Section 2.2 to replicate Hou and Hughes' experiments. The initial parameters for the numerical simulations fully coincided with those from the experimental tests. Hou and Hughes conducted experiments for intrinsic kinetic steam methane reforming for catalyst particles with no intra-particle diffusion limitation, and the catalyst particle size was about 0.15 mm. Moreover, the mass of the catalyst was 0.3 g, and the catalyst particles were diluted with inert particles of the same diameter.

A computational domain and mesh structure for a set of numerical calculations based on Hou and Hughes' experiments is presented in Fig. 2a. The computational domain fully replicates the experimental setup and consists of three zones: inlet (50 mm), catalyst (338 mm), and outlet (50 mm). The diameter of the tubular reformer is 10 mm. The constant temperature on the reformer wall was controlled by an electrical heater; thus, a constant wall temperature was set as a boundary condition for the reformer wall.

In the numerical model based on Hou and Hughes' experiments, the catalyst layer was modeled using a pseudo-homogeneous approach. The pseudo-homogeneous model calculates a momentum source term for the standard fluid flow equations. The source term is composed of two parts: a viscous loss term and an inertial loss term::

$$S_i = - \left(\frac{\mu}{\alpha} v_i + C_2 \frac{1}{2} \rho |v| v_i \right) \quad (21)$$

where α – permeability; C_2 – inertial resistance factor.

The values of the permeability (α) and the inertial resistance factor (C_2) were added to porous zone module in Fluent solver.

Fig. 2b depicts a good correlation between numerical results and experimental data, with less than 7% variation observed between them. Moreover, the temperature contour confirmed Hou and Hughes' assumption that the reactor can be considered isothermal because the temperature deviation inside the catalyst layer is less than 1 K. Therefore, it can be concluded that the numerical model provides high accuracy for predicting steam methane reforming parameters and can be used for further analysis of steam methane reforming over catalyst-coated hot wire.

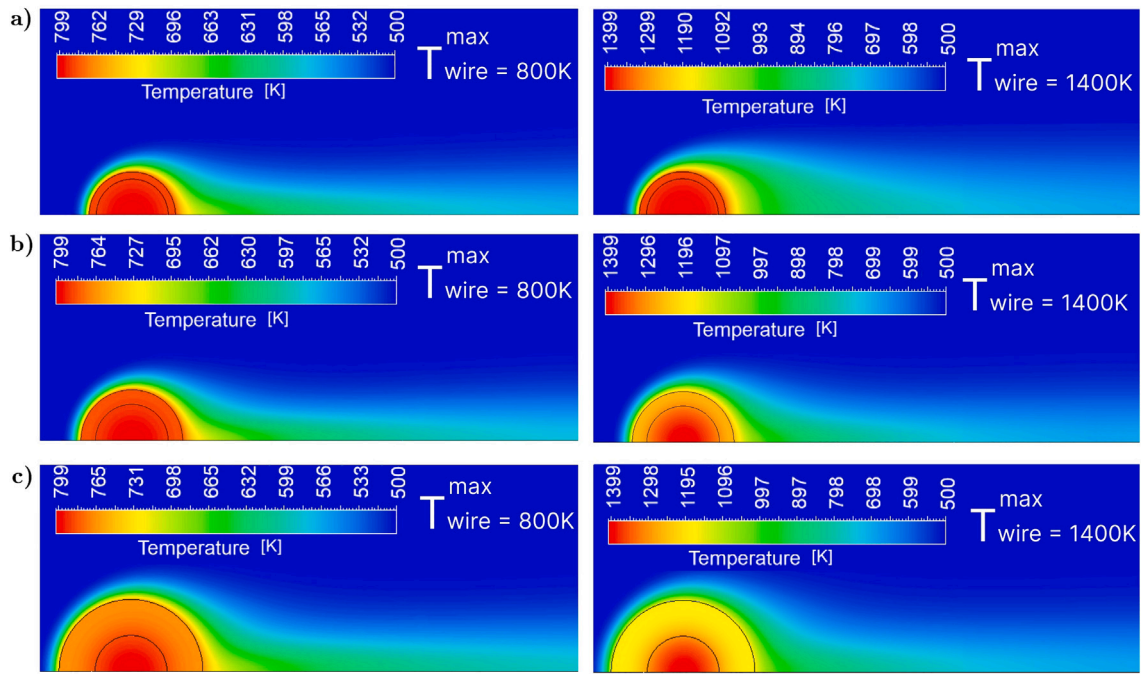


Fig. 3. The temperature contours at $Re=100$ for various catalyst layer thickness: (a) 1 mm; (b) 2 mm; (c) 5 mm.

3.2. Temperature

Steam methane reforming is a strongly endothermic process that consumes heat for chemical reactions. The endothermic effect (enthalpy of the SMR process) depends on the reaction rates of reactions (1)–(3): a higher reaction rate provides a higher endothermic effect. At the same time, an increase in temperature leads to an increase in the reaction rates of steam methane reforming reactions. In general, steam methane reforming reactions occur in the porous catalyst layer. It is expected that steam methane reforming has a significant effect on heat balance as well as temperature distribution in the catalyst-coated wire. Fig. 3 shows the temperature contours at $Re=100$ for various catalyst layer thicknesses: (a) 1 mm; (b) 2 mm; (c) 5 mm. The volume heat flux for steady-state conditions was set in ANSYS Fluent, and the maximum temperature of the wire was controlled.

Fig. 3 illustrates the temperature deviation in the wire, catalyst layer, and flow region for maximum wire temperatures of 800 K and 1400 K. This figure reveals that there is negligible temperature deviation in the wire and catalyst layer for a thickness of 1 mm. This can be explained by the following factors: (a) in this case, the catalyst layer can be considered a thermally thin layer with high thermal conductivity (characteristic length is 0.001 m, thermal conductivity is 18.9 W/(m·K)), and (b) the mass of catalyst in the layer is minimal, leading to minimal methane conversion and, as a result, minimal process enthalpy.

In the description, the thermally thin definition is used that means a catalyst is considered thermally thin when its internal thermal resistance to conduction is much smaller than the external thermal resistance to convection (or radiation). This means:

- The temperature within the material is nearly uniform (no significant spatial temperature gradients).
- The Biot number (Bi) is small (typically $Bi < 0.1$).
- The heat transfer is dominated by external convection/radiation, not internal conduction.

If $Bi < 0.1$, the object is thermally thin, meaning:

- The temperature gradient inside the material is negligible.

- The entire object heats or cools almost uniformly.

Fig. 3 also depicts that an increase in catalyst layer thickness increases the temperature deviation in both the catalyst layer and the wire. There are a few reasons for this observation. The first reason is that a thicker catalyst layer has higher thermal resistance to heat flux from the hot wire to the catalyst layer. At the same time, endothermic reactions are taking place in the volume of the porous catalyst layer. Therefore, extensive heat consumption by chemical reactions cannot be fully compensated by heat flux from the hot wire. The second reason is the dependence of process enthalpy on the mass of the catalyst. An increase in the mass of the catalyst leads to an increase in the amount of heat absorbed by the endothermic chemical reactions. For $T_{wire}^{max}=1400$ K, the heat consumed by the endothermic chemical reaction is significantly higher than the heat flux from the wire, leading to notable temperature deviation in the wire. The maximum temperature is observed at the center of the wire, with a decrease along the radial direction to the outer diameter.

The quantitative deviation of temperature in the wire, catalyst layer, and flow region along the axial direction at $Re=100$ for various catalyst layer thicknesses is shown in Fig. 4. This figure shows that for a catalyst layer thickness of 1 mm, the temperature deviation in the wire is minimal: for 800 K it is 4 K; for 1400 K it is 31 K. The temperature difference in the catalyst layer between the inner and outer diameters is also minimal and is only a few degrees for all temperatures. However, an increase in catalyst thickness and maximum wire temperature leads to an increase in temperature deviation. For a catalyst layer thickness of 5 mm and $T_{wire}^{max}=800$ K, the temperature difference in the wire is 12 K, while for $T_{wire}^{max}=1400$ K, the difference is 102 K. The maximum temperature variation for the catalyst layer is also observed for $T_{wire}^{max}=1400$ K with a catalyst layer thickness of 5 mm.

Fig. 4 presents another interesting observation. The temperature variation along the axial direction for both the catalyst layer and the wire exhibits a nearly symmetrical appearance for the front (left) and back (right) sides of the catalyst-coated wire. Specifically, the temperature variation within the catalyst layer at 5 mm thickness, with $T_{wire}^{max}=1400$ K, amounts to approximately 8 K. The primary cause of this deviation stems from heat transfer via convection, where the reaction

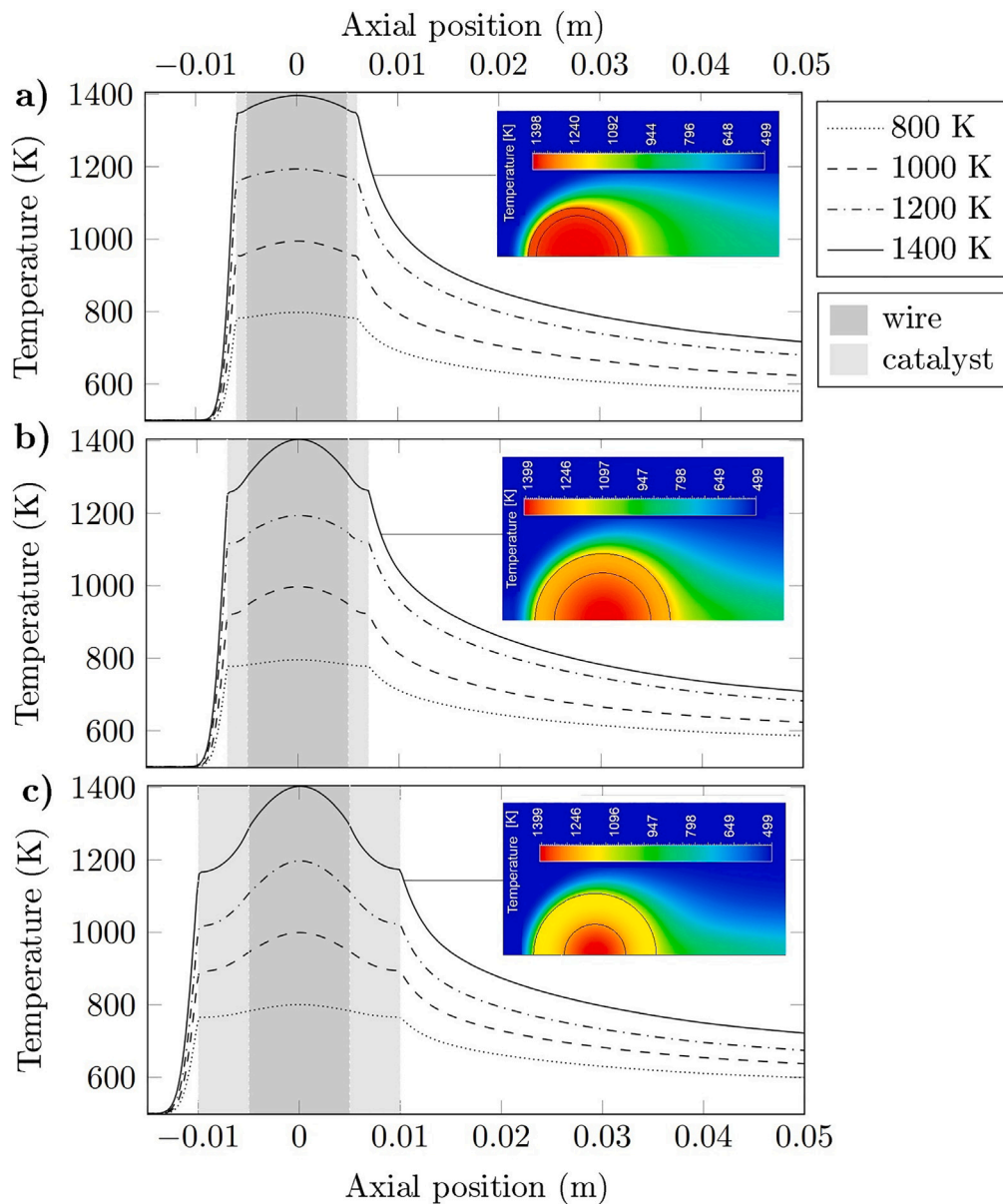


Fig. 4. The temperature distribution along axial line at $Re=100$ for various catalyst layer thickness: (a) 1 mm; (b) 2 mm; (c) 5 mm.

mixture remains colder on the left side and becomes heated on the right side. Simultaneously, this deviation indicates that the heat consumed by chemical reactions predominates within the catalyst-coated wire since transport phenomena are primarily governed by diffusion and thermal conductivity, whereas convection exerts a negligible influence on transport phenomena inside the catalyst layer.

As previously mentioned, the endothermic effect of the reaction demonstrates a nonlinear dependence on temperature. Consequently, the heat flow per running meter of the hot wire also varies nonlinearly with temperature. Fig. 5 illustrates the heat flow from the wire as a function of temperature and Reynolds number for various catalyst layer thicknesses: (a) 1 mm, (b) 2 mm, and (c) 5 mm. This figure reveals that increasing the catalyst layer thickness results in a nonlinear rise in the heat flow, even though a thicker layer enlarges the outer diameter of the catalyst-coated wire. A key factor contributing to this phenomenon is the intra-particle diffusion limitation within the catalyst layer. Additionally, Fig. 5 highlights that elevating the Reynolds number substantially enhances the heat flow for thicknesses of 2 mm and 5 mm, chiefly attributable to augmented mass transfer from the catalyst surface to the reaction mixture.

Moreover, it should be noted that an increase in catalyst layer thickness increases the heat transfer surface. The outer surface area of a catalyst layer of 2 mm and 5 mm is 1.19 and 1.86 times greater, respectively, than that of a catalyst layer of 1 mm. To assess the effect of catalyst layer thickness on heat flow per 1 m^2 of outer surface (Q'') for one running meter of wire, we calculated it and plotted the results in Fig. 6.

Fig. 6 shows important results. For this figure, it can be assumed that the heat transferred for the change of enthalpy per 1 m^2 of catalyst surface for all three cases is similar. Therefore, the increase in heat flow with an increase in catalyst thickness should be driven by the enthalpy of the steam methane reforming process. However, we can see from Fig. 6 a non-linear increase in heat flow for catalyst layer thicknesses of 1 mm, 2 mm, and 5 mm. A notable increase in Q'' can be observed when the thickness changes from 1 mm to 2 mm. However, if the catalyst layer thickness increases from 2 mm to 5 mm, a negligible increase in heat flow can be observed. The main reason for this observation is that an increase in catalyst layer thickness does not increase the enthalpy of the steam methane reforming process and, as a result, does not enhance

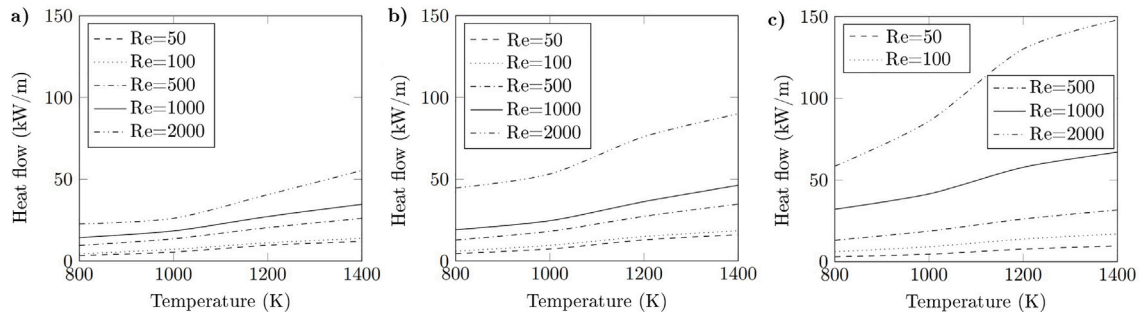


Fig. 5. Heat flow per running meter of wire as a function of temperature and Reynolds number for various catalyst layer thickness: (a) 1 mm; (b) 2 mm; (c) 5 mm.

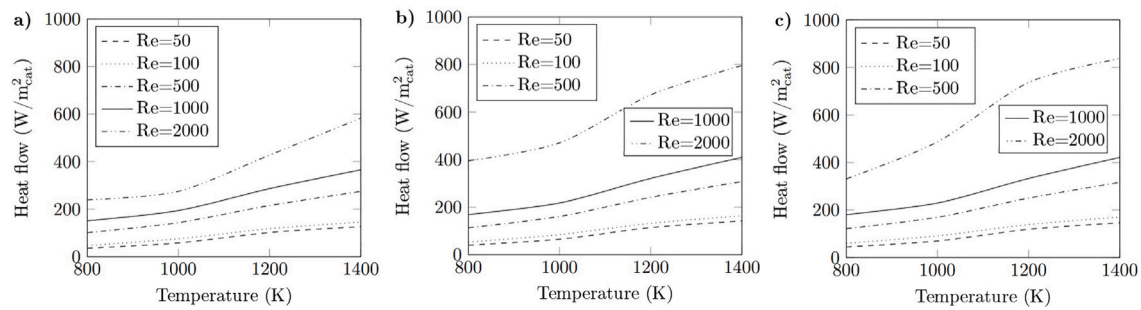


Fig. 6. Heat flow per running meter of wire as a function of temperature and Reynolds number for various catalyst layer thickness: (a) 1 mm; (b) 2 mm; (c) 5 mm.

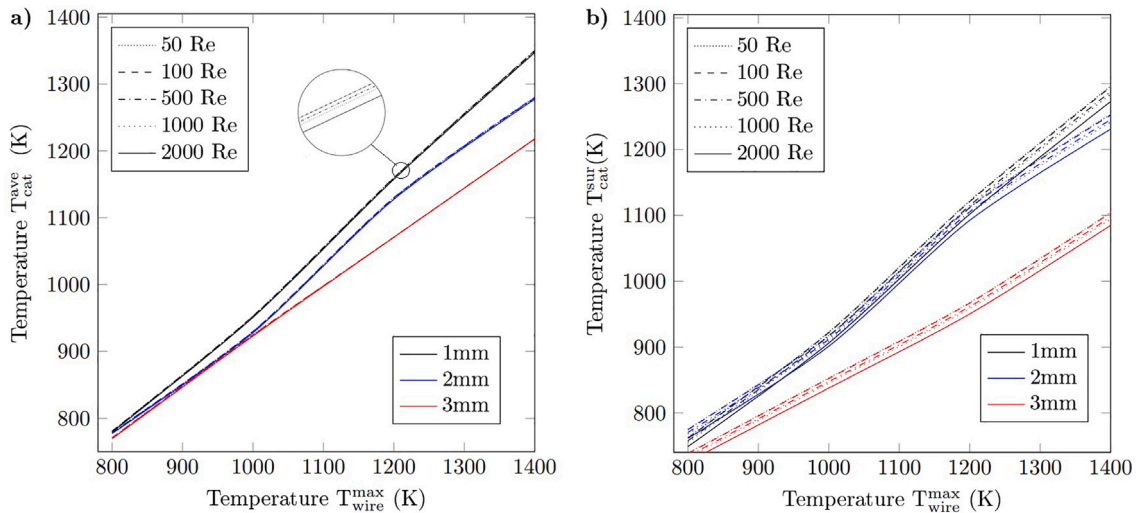


Fig. 7. Dependence of catalyst average temperature (a) and catalyst surface temperature (b) on maximal wire temperature for the different Re and catalyst layer thickness of 1 mm, 2 mm, and 5 mm.

methane conversion due to intra-layer diffusion limitations discussed in the author's previous paper [45].

Another reason for the non-linear change in enthalpy of the steam methane reforming process with respect to catalyst layer thickness is the average catalyst temperature and the temperature of the catalyst on the outer surface. Fig. 7 shows the dependence of the average catalyst temperature (a) and the catalyst surface temperature (b) on maximal wire temperature for different Reynolds numbers (Re) and catalyst layer thicknesses of 1 mm, 2 mm, and 5 mm.

Fig. 7a shows that for a thin catalyst layer, the average catalyst temperature is close to the maximal wire temperature because heat from the wire is transferred via thermal conduction, and the catalyst layer of 1 mm can be considered thermally thin. At the same time, an increase in catalyst layer thickness to 5 mm results in a difference between maximal wire temperature and average catalyst temperature of up to 190 K for $T_{wire}^{max} = 1400$ K. However, a decrease in maximal wire temperature reduces this difference due to a lower reaction rate at lower temperatures. Moreover, Fig. 7b shows that the difference

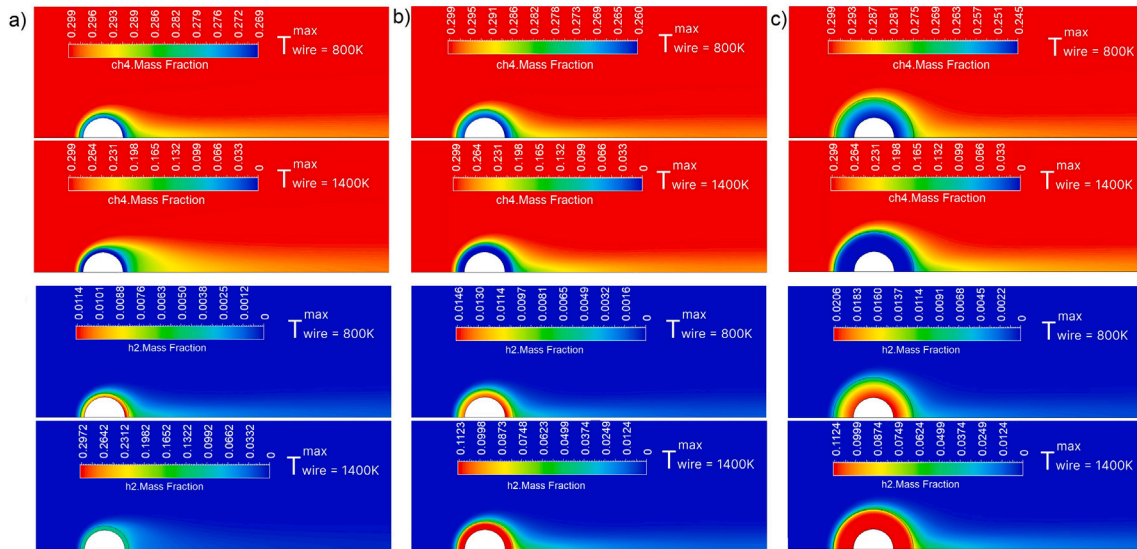


Fig. 8. The mass fraction of H_2 and CH_4 for various catalyst layer thickness at $T_{\text{wire}}^{\text{max}}=800$ K; 1400 K and $Re=100$: (a) 1 mm; (b) 2 mm; (c) 5 mm; (upper contours – CH_4 mass fraction; bottom contours – H_2 mass fraction).

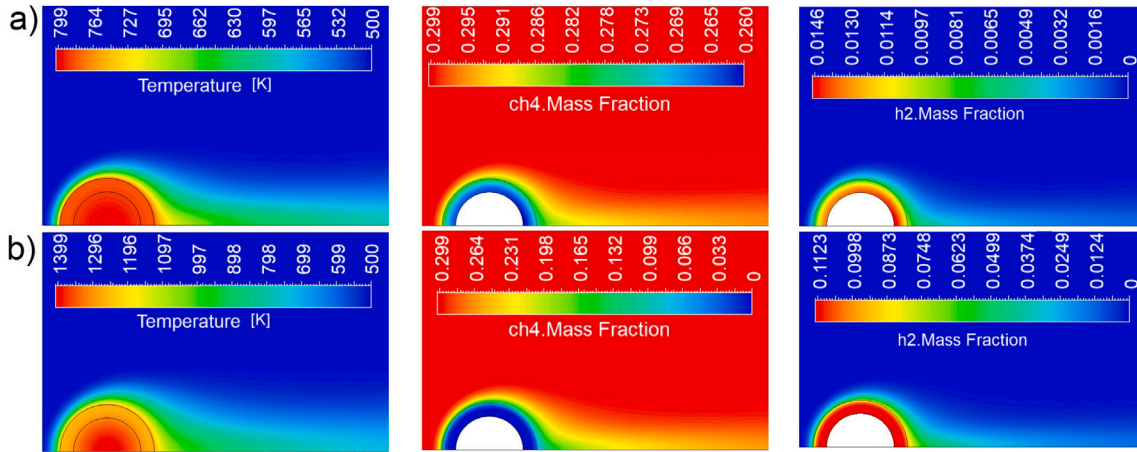


Fig. 9. The contours of temperature, mass fraction of H_2 and CH_4 at $T_{\text{wire}}^{\text{max}}=800$ K (a); $T_{\text{wire}}^{\text{max}}=1400$ K (b) and $Re=100$ for the catalyst layer thickness of 2 mm.

between maximal wire temperature and catalyst surface temperature is significant and can reach up to 320 K for $T_{\text{wire}}^{\text{max}} = 1400$ K and a thickness of 5 mm.

3.3. Mass fraction

Operational parameters (such as temperature and Reynolds number) and design parameters (like catalyst layer thickness) profoundly impact the composition of the reaction mixture. Fig. 8 displays the mass fractions of H_2 and CH_4 for diverse catalyst layer thicknesses at $T_{\text{wire}}^{\text{max}}$ values of 800 K and 1400 K. This figure distinctly shows that chemical reactions transpire within the catalyst layer. For $T_{\text{wire}}^{\text{max}}=1400$ K, the CH_4 mass fraction reaches a minimum across all thicknesses, indicating near-complete conversion of methane into synthesis gas. A thin layer of methane proximal to the catalyst's surface is discernible for thicknesses of 2 mm and 5 mm; notably, this layer appears thicker on the left side compared to the right. This disparity can be attributed to convective transport occurring near the surface. Fig. 8 further demonstrates that maximum H_2 production occurs at $T_{\text{wire}}^{\text{max}} = 1400$ K, whereas this value is lowest at $T_{\text{wire}}^{\text{max}} = 800$ K. The contours of H_2 and CH_4 mass fractions are intrinsically connected to those of temperature, as depicted in Fig. 9. Higher temperatures correlate with elevated H_2 mass fractions and

diminished CH_4 mass fraction. Specifically, at $T_{\text{wire}}^{\text{max}} = 1400$ K, the mean CH_4 mass fraction drops to 0.02, confirming that virtually complete methane conversion is achievable under these conditions.

The quantitative comparison of CH_4 and H_2 mass fractions is presented in Fig. 10. Fig. 10 shows the average mass fractions of methane and hydrogen as functions of temperature and Reynolds number for various catalyst layer thicknesses: (a) 1 mm, (b) 2 mm, and (c) 5 mm. This figure reveals that an increase in Reynolds number leads to an increase in H_2 mass fraction and a decrease in CH_4 mass fraction. One explanation for this observation is that higher Reynolds numbers result in higher mass transfer coefficients from the catalyst surface to the reaction mixture. However, a substantial increase in the Reynolds number causes only minor changes in CH_4 and H_2 mass fractions. This finding confirms that within the catalyst layer, transport phenomena are predominantly driven by diffusion, and higher Reynolds numbers only affect the mass transfer coefficient from the catalyst surface.

Furthermore, Fig. 10 illustrates the effect of temperature on CH_4 and H_2 mass fractions. It shows a nonlinear dependence of CH_4 and H_2 mass fractions on temperature due to the following factors: (a) reaction rates depend nonlinearly on temperature; (b) at temperatures above a certain threshold, methane conversion approaches unity; and (c) higher H_2 mass fraction exacerbates intra-particle diffusion limitations.

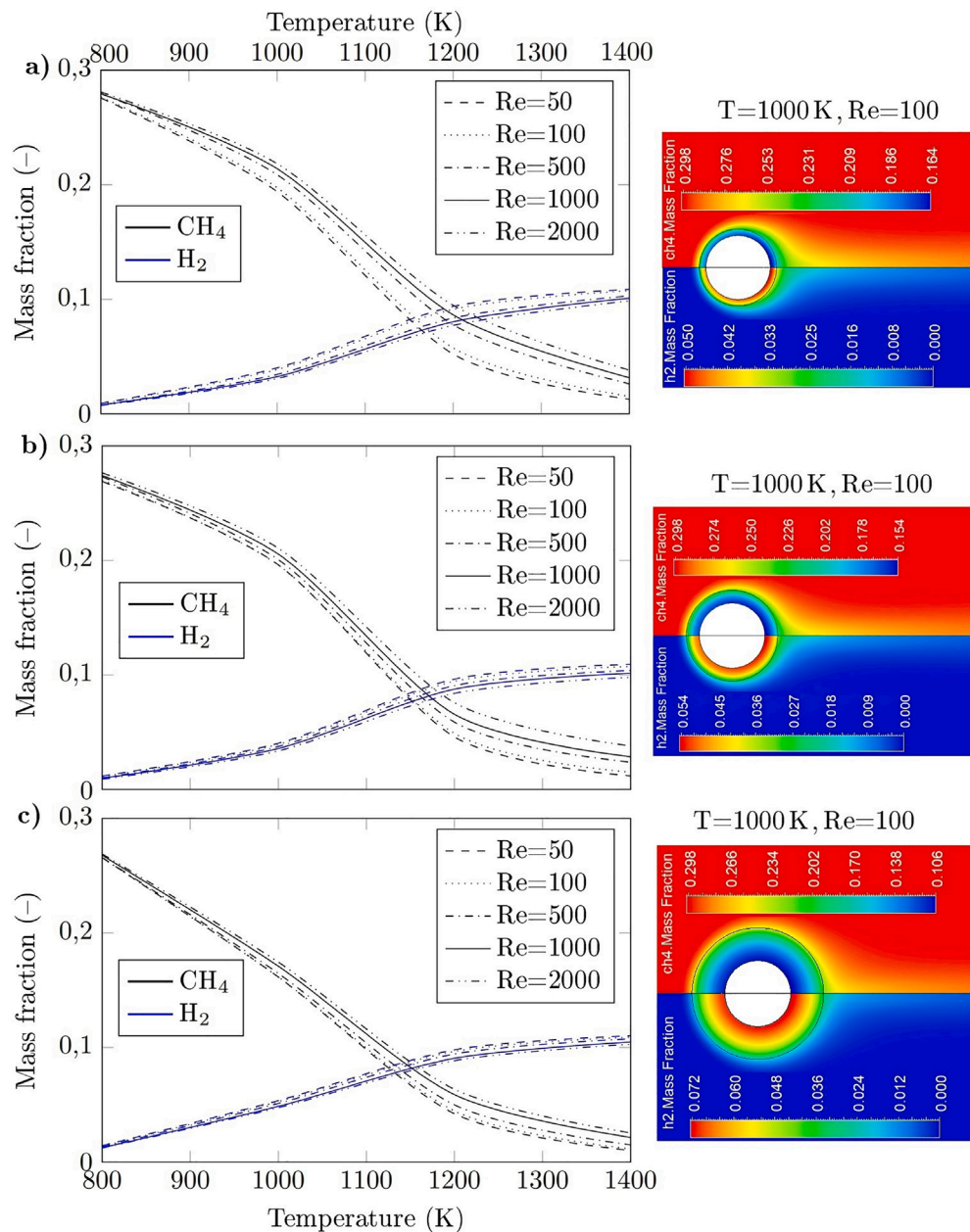


Fig. 10. The average mass fraction of methane and hydrogen as a function of temperature and Reynolds number for various catalyst layer thickness: (a) 1 mm; (b) 2 mm; (c) 5 mm.

Carbon formation is another factor that needs to be considered in steam methane reforming. There is sufficient information in research papers indicating that carbon formation can be omitted for steam-to-methane ratios above 1.0 over coke-resistant catalysts [46, 47]. Furthermore, we analyzed changes in Gibbs free energy for various possible chemical reactions in steam methane reforming, including carbon formation. Fig. 11 shows that in the temperature range above 1000 K, the chemical reaction of steam methane reforming is more favorable from a thermodynamic point of view than reactions leading to carbon formation, such as methane cracking and the Boudouard reaction.

Therefore, it can be concluded that in the present study, carbon formation and deposition have a negligible effect on material and energy balances. As a result, the chemical kinetic model considers only the reaction of steam methane reforming coupled with the water-gas shift reaction.

4. Conclusion

This study explored the potential of steam methane reforming (SMR) using renewable electricity as a promising method for low-carbon hydrogen production. Specifically, this paper focused on SMR over a catalyst-coated wire heated by Joule heating, investigating the heat and mass transfer within the catalyst-wire system. The numerical model was validated against experimental data with a good correlation with maximal deviation about 4.7%.

The analysis of the temperature deviation in the wire, catalyst layer, and flow region for the different maximum wire temperatures showed that at a catalyst layer thickness of 1 mm, the temperature deviations in the wire and catalyst layer were minimal. This can be attributed to the high thermal conductivity of the catalyst layer and its minimal mass, resulting in limited methane conversion and consequently minimal process enthalpy. Quantitatively, the temperature deviation in the wire for a 1 mm catalyst layer was only 4 K at 800 K and 31 K at 1400 K.

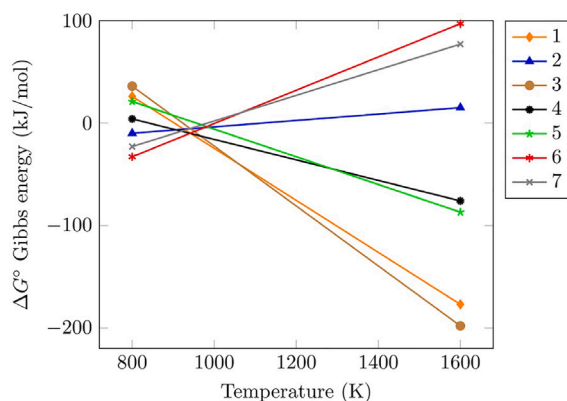


Fig. 11. The change in Gibbs free energy [48].

The heat flow from the wire as a function of temperature and Reynolds number for different catalyst layer thicknesses was analyzed. Thicker catalyst layers led to a nonlinear increase in heat flow despite their larger outer diameters. This phenomenon is primarily due to intra-particle diffusion limitations within the catalyst layer. Elevating the Reynolds number notably improved the heat flow for 2 mm and 5 mm thicknesses, mainly because of enhanced mass transfer from the catalyst surface to the reaction mixture.

The analysis of the effect of operational and design parameters on mass balance showed that higher temperatures lead to increased H_2 mass fractions and decreased CH_4 mass fractions, indicating enhanced methane conversion. At $T_{wire}^{max} = 1400$ K, nearly complete methane conversion is achieved, with the mean CH_4 mass fraction dropping to 0.02. It was established that increasing the Reynolds number raises H_2 mass fractions and reduces CH_4 mass fractions due to improved mass transfer from the catalyst surface to the reaction mixture. However, substantial increases in the Reynolds number cause only slight changes in CH_4 and H_2 mass fractions, suggesting that diffusion dominates transport phenomena within the catalyst layer. The discussion of potential carbon formation was provided clarifying that the given initial parameters is not favorable for the carbon formation.

CRediT authorship contribution statement

Dmitry Pashchenko: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Viacheslav Papkov:** Writing – review & editing, Visualization, Validation, Software, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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No.	Reaction	ΔH_{298} , (kJ/mol)
Methane oxidation		
1	$CH_4 + H_2O \rightleftharpoons CO + 3H_2$	206.4
2	$CO + H_2O \rightleftharpoons CO_2 + H_2$	-41.1
3	$CH_4 + CO_2 \rightleftharpoons 2H_2 + 2CO$	247.3
Carbon formation		
4	$CH_4 \rightleftharpoons C + 2H_2$	74.8
5	$2CO \rightleftharpoons C + CO_2$	-173.3
6	$CO + H_2 \rightleftharpoons C + H_2O$	-90.13
7	$CO_2 + 2H_2 \rightleftharpoons C + 2H_2O$	-131.3

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